Remnant Fit 3D

Project report, version 2 · Manycore Aholo Global 3D AI Hackathon · Team "Design Master" (Alan De Bortolo, Design Master Solutions LLC)

Play: remnant-fit-3d.designmaster.workers.dev · Code: github.com/DesignMaster-Solutions/remnant-fit-3d · Demo video: see the submission form

1. The problem

Stone shops cut kitchen countertops from big slabs. The leftovers (offcuts, or "remnants") pile up in the yard, and many end up in a skip because a buyer cannot picture an odd-shaped piece of marble on their own counter. I build software for stone fabricators, and "will that piece fit my kitchen?" is the question their customers ask most.

Remnant Fit 3D turns that question into a browser game: five kitchens, a yard full of offcuts, and a foreman whose plan you try to beat. Everything you place is stone that did not go to the skip.

2. What changed since Version 1

Area	Version 1 (30 Oct)	Version 2 (this submission)
Lux3D use	4 text-to-3D tasks	17 tasks: text-to-3D, image generation, image-to-3D and material transfer; 314 credits
Levels	1 kitchen	5 jobs, each a different generated kitchen: straight run, galley, U-shape, L with sink, island with cooktop
Goal	Free puzzle, no end	Beat the foreman: a planner solves each level at load time from the mesh; stars against his score; running total of kilograms kept from the skip
Feel	Silent	Synthesised stone knocks, saw whine, chimes; slab lifts, drops with a bounce, dust and a small camera shake; orange tiles show what the saw takes; phone vibration where supported
Onboarding	One hint line	A three-line intro card, a coach line that changes after the first move, toasts for saw cuts and blocked drops
Phone	Played, small	Camera fits counter and yard between header and footer on any screen; double tap turns a slab
Level reading	Ray cast per cell, 11.8 s for the L kitchen in node	Top-down triangle rasteriser, same grid, 10 ms
Tests	3 unit tests	15 node tests (rules, foreman, every level solvable) and an in-page touch test

3. How Aholo Lux3D is used

Every 3D object in the game is a Lux3D generation, made by `lux3d/gen.py` through the Lux3D API. The script creates each task, waits for tasks it depends on (an image before its image-to-3D, a swatch before its material transfer), polls, downloads, and logs every event to `lux3d/gen.log`. Task list and credits below come from the API's own task list.

Asset	Lux3D capability	Task ID	Credits	Time	Used as
remnant-a, -b, -c	Text-to-3D G1	4502786, 4502793, 4502801	60	~3 min	Calacatta, Absolute black, travertine offcuts (Version 1)
kitchen	Text-to-3D G1	4502808	20	3.7 min	Job 4, L with sink (Version 1)
kitchen-run	Text-to-3D G1	4591991	20	5.4 min	Job 1, straight run
kitchen-galley	Text-to-3D G1	4591998	20	4.3 min	Job 2, galley
kitchen-u	Text-to-3D G1	4592004	20	4.7 min	Job 3, U-shape
kitchen-island	Text-to-3D G1	4592013	20	5.6 min	Job 5, island with cooktop
kitchen-u-ref	Multimodal image	4592024	12	27 s	Input photo for image-to-3D
kitchen-u-img	Image-to-3D G1	4592044	20	5.4 min	Comparison against text-to-3D (section 4)
remnant-d, -e, -f	Text-to-3D G1	4592094, 4592155, 4592160	60	~3.2 min	Verde quartzite, terrazzo, Emperador offcuts
swatch-azul, swatch-rosa	Multimodal image	4592104, 4592200	24	32 to 50 s	Stone swatches for material transfer
remnant-azul, -rosa	Material transfer v3.0	4592170, 4592219	38	3.3 to 5 min	Azul Bahia and Rosa Portogallo offcuts on the Calacatta mesh
17 tasks			**314**		**0 failed**



Job 1: straight run



Job 2: galley



Job 3: U-shape



Job 5: island



Verde quartzite



Terrazzo strip



Emperador triangle



Absolute black L

Raw renders of the downloaded meshes (Blender, `report/render.py`), before any processing.

4. Key asset iterations

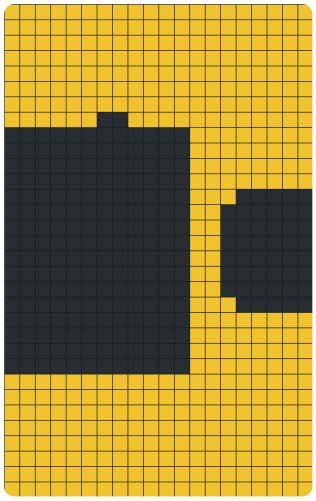
Same U brief, two routes: image-to-3D against text-to-3D



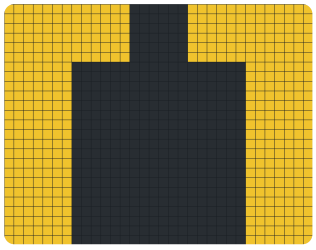
Aholo image (task 4592024)



Image-to-3D from it



What the game reads: false gap, ragged sink



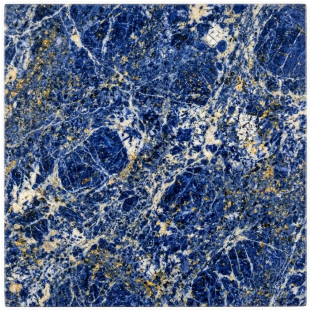
Text-to-3D U: clean, 422 cells

The image route gave the nicer-looking model, but read from above it had a one-cell gap across a run and a jagged sink outline. A puzzle on a 10 cm grid needs clean edges, so job 3 uses the text-to-3D kitchen and the image-to-3D one stays in the repository as the comparison.

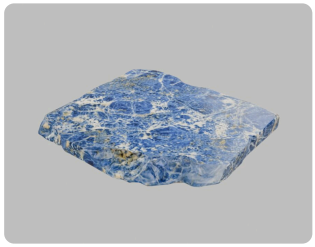
Material transfer: new stones without new shapes



Carrier: Calacatta offcut



Aholo swatch



Azul Bahia result



Rosa Portogallo result

Material transfer kept the exact outline of the Calacatta mesh and repainted it from a generated swatch. In a real yard the same cut shape shows up in many stones, and this is how the game adds stones cheaply.

What did not come out as asked, and what I did

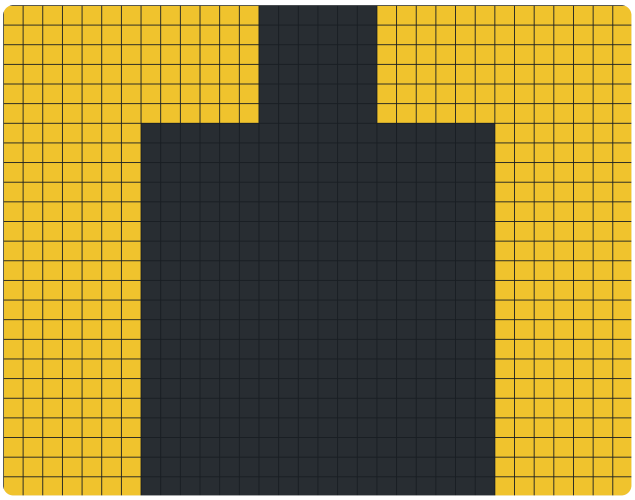
Asset	Asked for	Got	Decision
remnant-e	Terrazzo strip with a rectangular notch	A straight strip, the notch only a shallow recess	Kept: long strips are what galley runs need
kitchen-run	Plain straight run, bare top	Bar handles rising to top height and small recesses	The reader drops anything one cell wide (a 2x2 opening) and patches under 20 cells
kitchen-u-img	U kitchen from a photo	False gap and ragged sink in the top view	Comparison only, see above

5. The key idea: the level is read from the generated mesh

There is no hand-made level data. On load the game stands each kitchen at 0.9 m, scales its longest run to a stated length, and rasterises every triangle seen from above onto a 10 cm grid, keeping the highest surface per cell. Cells at the top level are counter; the sink, the cooktop hole and the floor drop out. Slabs are laid flat and read the same way. The fitting rules live in `public/fit.js`, the reader in `public/mesh-grid.js`, and the node tests load the same GLB files through the same code, so the game and the tests see one truth. A new Lux3D kitchen is a new level: `public/levels.js` only names the model, its length and the yard stock.



Lux3D kitchen-u.glb



What the game reads from it (yellow = counter)

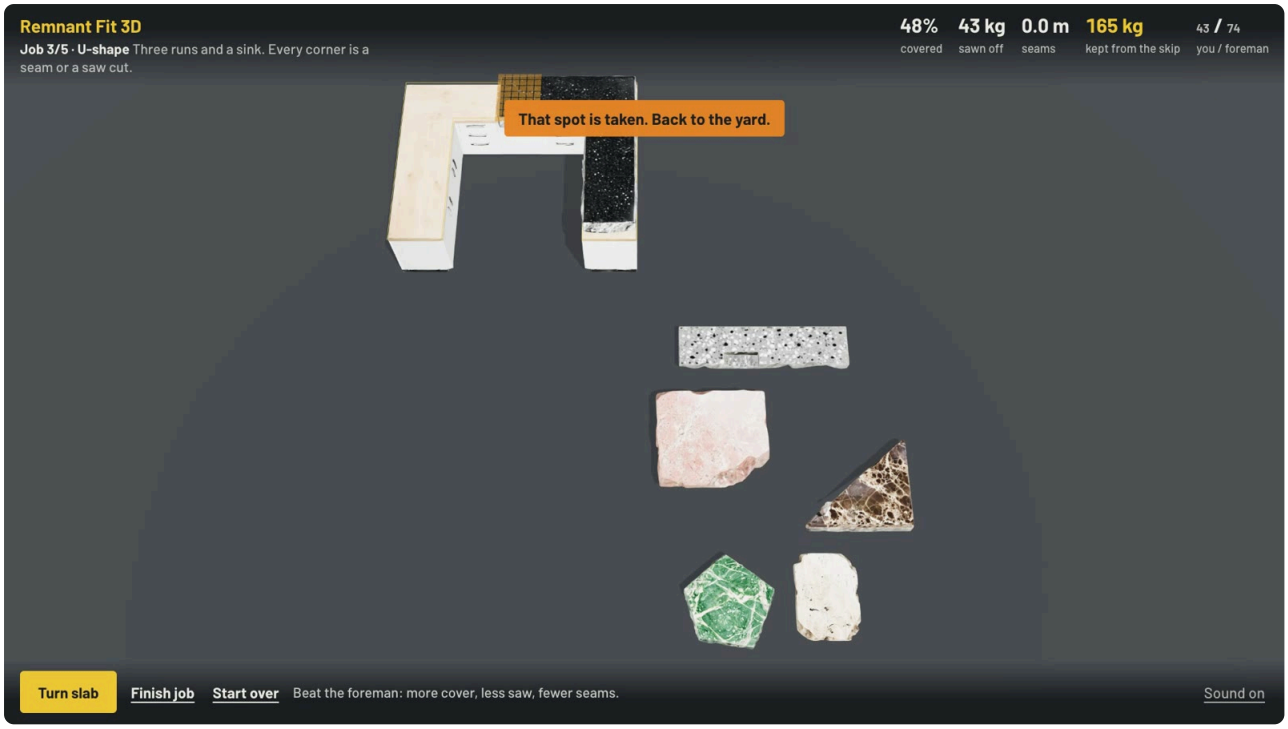
6. Gameplay

Each job is a client brief and a yard. Drag offcuts onto the yellow outline, turn them (button, R key, or double tap), and finish the job when you are happy. Two slabs cannot overlap; overhang past the counter or over a hole is legal but sawn off and shown in orange; every edge where two slabs meet is a seam. Score = cover % minus 10 per m^2 sawn off minus 5 per metre of seam.

The foreman is a greedy planner (`foreman()` in `public/fit.js`): biggest slab first, each at the spot and turn that adds the most score. His score on the current level is the bar: 70% of it earns one star, 90% two, matching or beating him three. Finishing reports kilograms kept from the skip, kilograms sawn off, and the new stone that would still be needed, and adds to a running total across the five kitchens. Weights use 3 cm stone at 2.7 t/m^3 (81 kg per m^2), an assumption stated in the game, not a measured yard figure.



Job 3 on a phone screen (iOS Simulator, WebKit)



A frame of the demo film: the real game, played through its own pointer handlers.

Balance, measured

Job	Kitchen	Counter	Foreman cover	Sawn off	Seams	Foreman score
1	Straight run	1.70 m^2	100%	0.95 m^2	1.4 m	84
2	Galley	3.18 m^2	93%	1.09 m^2	0.0 m	82
3	U-shape	4.22 m^2	93%	1.68 m^2	0.4 m	74
4	L with sink	5.83 m^2	85%	1.29 m^2	0.4 m	70
5	Island with cooktop	3.49 m^2	97%	2.43 m^2	0.8 m	69

Printed by `npm test`. I tuned the yard sizes until the first job was forgiving and the jobs got harder in order; the foreman is greedy, so a careful player can beat him.

7. Testing

- **15 node tests** (`npm test`): rotation, overlap, scoring, the foreman (no overlaps, turns when it must, leaves useless slabs in the yard), stars, job result, the handle filter, every level read from its real GLB and covered at least 80% by the foreman, and the v2 reader giving the same grids as the v1 ray caster (583 and 197 cells). Each new test was seen failing before the code that passes it.
- **In-page touch test** (`?selftest`, `public/selftest.js`) through the game's own pointer handlers, run in WebKit on the iOS Simulator: drag places a slab, a slab dropped on another goes back to the yard, turn, start over, the foreman's plan laid by hand scores 84 vs 84 and earns three stars, next job loads.
- The demo film is rendered by HyperFrames from the real game page; a director script plays it frame by frame through the same handlers (`video/director.js`), and the soundtrack is the game's own synth rendered offline from the recorded events (`video/soundtrack.mjs`).

8. Why it matters

Each counter covered with offcuts is stone that is sold instead of thrown away. The pipeline in this game (describe or photograph an offcut, generate it with Lux3D, read its outline from the mesh, try it on a generated kitchen) is the tool a stone shop needs to put its yard online and let customers try a piece on their own kitchen before they buy it.

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